

DAY 21

HYPOTHESIS TESTING IN DATA SCIENCE

Detailed practical guide using Python, NumPy, Pandas, and SciPy

Prepared for Muhammad Ahmad

60-Day Data Science Learning Journey

Learning outcomes

- Formulate null and alternative hypotheses.
- Interpret significance levels, p-values, and test statistics.
- Choose one-sample, independent, and paired t-tests.
- Run tests with SciPy and explain results in plain English.
- Build a Student Group Comparison Analyzer.

How to use this guide

Read the concept, run the example, compare your output, and attempt the tasks before reading the solutions. Small differences in decimal places may occur across software versions.

Environment setup: install the libraries in a terminal or notebook.

```
python -m pip install numpy pandas scipy
```

1. What is hypothesis testing?

Hypothesis testing is a statistical procedure used to evaluate whether sample data provides sufficient evidence against a claim about a population. It does not prove a claim with absolute certainty.

Example: a claim about student marks

Suppose a university claims the population mean mark is 75. A sample of students is used to test whether the population mean differs from 75.

Null hypothesis: $H_0: \mu = 75$
Alternative hypothesis: $H_1: \mu \neq 75$

μ (mu) represents the population mean. H_0 is the null hypothesis, often the baseline claim. H_1 , also called H_a , is the alternative hypothesis.

Key vocabulary

Term	Meaning
Population	The full group you want to understand.
Sample	The observations collected from the population.
Parameter	A population quantity, such as the true population mean μ .
Statistic	A value calculated from a sample, such as sample mean $\bar{x}$.
Test statistic	A standardized value calculated by a test, such as a t-statistic.
Significance level (alpha)	A threshold chosen before the test, often 0.05.
P-value	Under H_0 and the test assumptions, the probability of a result at least as extreme as the observed one.

2. Hypothesis-testing workflow

- State the research question and identify the population parameter.
- Write H_0 and H_1 . Decide whether the test is two-tailed, right-tailed, or left-tailed.
- Choose alpha before interpreting results, often 0.05.
- Select a suitable test and check its assumptions.
- Collect data and calculate the test statistic and p-value.
- Compare the p-value with alpha and apply the decision rule.
- Explain the conclusion in context, including limitations and practical importance.

3. Significance level and p-value

Condition	Decision	Interpretation
p-value < alpha	Reject H0	Evidence against H0 under the selected test and assumptions.
p-value >= alpha	Fail to reject H0	Insufficient evidence to reject H0 at this threshold.

Important: a p-value is not the probability that H0 is true. A p-value of 0.03 does not mean there is a 3% chance that H0 is true. Failing to reject H0 does not prove H0 is true.

Statistical significance and practical importance are different. A tiny effect can be statistically significant in a large sample. Examine effect size, confidence intervals, study design, and context as well as the p-value.

4. One-tailed and two-tailed tests

Test	Alternative hypothesis	Question
Two-tailed	H1: $\mu \neq 75$	Is the mean different from 75 in either direction?
Right-tailed	H1: $\mu > 75$	Is the mean greater than 75?
Left-tailed	H1: $\mu < 75$	Is the mean less than 75?

Choose the test direction from the research question before looking at the results. Do not switch to a one-tailed test just to obtain a smaller p-value.

5. One-sample t-test

Use a one-sample t-test to compare a sample mean with a fixed reference value when the population standard deviation is unknown. The test estimates variability using the sample standard deviation.

Formula: $t = (\text{sample mean} - \text{reference mean}) / (\text{sample standard deviation} / \text{square root of } n)$

The denominator is the estimated standard error. The t-test assumes independent observations. With a small sample, the data should not have severe outliers or strong departures from normality.

Complete example

```
import numpy as np
from scipy import stats

marks = np.array([72, 75, 78, 80, 82, 85, 88, 90])
claimed_mean = 75

result = stats.ttest_1samp(
    marks,
    popmean=claimed_mean,
    alternative="two-sided"
)

print("Sample mean:", np.mean(marks))
print("T-statistic:", result.statistic)
print("P-value:", result.pvalue)

alpha = 0.05
if result.pvalue < alpha:
    print("Reject the null hypothesis.")
else:
    print("Fail to reject the null hypothesis.")
```

Line-by-line explanation

- import numpy as np imports NumPy with the alias np.

- `from scipy import stats` imports SciPy's statistical functions.
- `np.array([...])` stores eight numerical observations.
- `claimed_mean = 75` stores the reference mean under H_0 .
- `stats.ttest_1samp(...)` performs a one-sample t-test.
- `popmean=claimed_mean` passes the reference value.
- `alternative='two-sided'` tests for a difference in either direction.
- `np.mean(marks)` calculates the sample average.
- `result.statistic` returns the t-statistic.
- `result.pvalue` returns the p-value.
- `alpha = 0.05` sets the decision threshold.
- The if/else statement applies the decision rule.

Expected output

```
Sample mean: 81.25
T-statistic: approximately 3.6
P-value: approximately 0.01
Reject the null hypothesis.
```

The sample mean is 81.25. The test provides evidence that the population mean differs from 75 at the 5% significance level. The test alone does not explain why the difference exists.

6. Independent-samples and Welch's t-test

An independent-samples test compares two separate groups, such as two different classes. Observations should be independent within and across groups. Welch's t-test does not require equal population variances and is selected in SciPy with `equal_var=False`.

```
from scipy import stats

group_a = [72, 75, 78, 80, 82]
group_b = [81, 83, 85, 87, 89]

result = stats.ttest_ind(
    group_a, group_b,
    equal_var=False,
    alternative="two-sided"
)

print("Group A mean:", sum(group_a) / len(group_a))
print("Group B mean:", sum(group_b) / len(group_b))
print("T-statistic:", result.statistic)
print("P-value:", result.pvalue)
```

Line-by-line explanation

- `group_a` and `group_b` store marks from separate groups.
- `stats.ttest_ind` compares the two group means.
- `equal_var=False` selects Welch's t-test, allowing unequal variances.
- `alternative='two-sided'` tests for a difference in either direction.
- The mean calculations summarize the observed groups.
- The p-value is interpreted against the preselected significance level.

Do not use an independent test for repeated measurements on the same students. A paired test may be suitable instead.

7. Paired t-test

Use a paired t-test when each observation in one condition corresponds to an observation in another condition, such as the same students' marks before and after training.

```
from scipy import stats
```

```

before = [60, 65, 70, 72, 68, 75]
after  = [68, 70, 74, 78, 75, 80]

result = stats.ttest_rel(
    before, after,
    alternative="two-sided"
)

print("Mean before:", sum(before) / len(before))
print("Mean after:", sum(after) / len(after))
print("P-value:", result.pvalue)

```

Line-by-line explanation

- before stores the first measurement for each participant.
- after stores the matching second measurement in the same order.
- stats.ttest_rel tests whether the mean paired difference is zero.
- The order of the two lists must match participant by participant.
- The paired differences should be independent across pairs and reasonably compatible with a normal distribution, especially in a small sample.

8. Z-test vs. t-test

Method	Typical use	Key point
Z-test for a mean	Population standard deviation known, with appropriate distribution assumptions.	Uses the standard normal reference distribution.
One-sample t-test	Population standard deviation unknown.	Uses a t distribution with $n - 1$ degrees of freedom.
Welch's t-test	Two independent groups; variances may differ.	Does not pool group variances.
Paired t-test	Matched observations or repeated measurements.	Tests the mean paired difference.

The population standard deviation is often unknown, so t-tests are common in real data analysis. Always consider the research question, independence, outliers, and data quality.

9. Assumptions and common mistakes

Assumptions to consider

- Independence: observations or paired differences should be independent according to the study design.
- The outcome should be numerical for the mean-based tests described here.
- For small samples, inspect severe outliers and strong non-normality in the relevant data or paired differences.
- Independent tests require separate independent groups; paired tests require matched observations.
- Biased sampling, confounding, or poor assignment can limit the conclusion.

Mistakes to avoid

- Treating $p < 0.05$ as proof of a hypothesis.
- Treating $p \geq 0.05$ as proof that no difference exists.
- Choosing the test direction after seeing results.
- Using an independent test for paired before-and-after measurements.
- Ignoring effect size, confidence intervals, sample size, or data quality.

-
- Reporting only significant results after testing many outcomes without considering multiple testing.
 - Claiming causation from a test when the study design does not support it.

10. Practice tasks

Try these tasks before reading the solutions in the next section.

Task 1: Sample mean

Use NumPy to calculate the mean of [60, 70, 80, 90, 100].

Task 2: One-sample t-test

Test whether [72, 75, 78, 80, 82, 85] differs from a reference mean of 75.

Task 3: Independent groups

Compare [70, 72, 74, 76] and [80, 82, 84, 86] with Welch's t-test.

Task 4: Interpret a p-value

A test gives $p = 0.02$ and $\alpha = 0.05$. State the decision accurately.

Task 5: Choose a test

The same students are measured before and after training. Which test should you consider?

Task 6: One-tailed hypothesis

Write the alternative hypothesis for the claim that the mean score is greater than 75.

Task 7: Independent vs paired

Explain why the same students' before-and-after marks are paired.

Task 8: Type I and Type II errors

Define both errors for a test about whether a teaching method changes average marks.

Task 9: Assumptions

Name two assumptions or data issues to check before interpreting a t-test.

Task 10: Practical significance

Why might a statistically significant result not be practically important?

11. Complete practice-task solutions

Solution 1: Sample mean

```
import numpy as np
marks = np.array([60, 70, 80, 90, 100])
print(np.mean(marks))
# Output: 80.0
```

The values sum to 400. Dividing by five observations gives a mean of 80.

Solution 2: One-sample t-test

```
from scipy import stats
marks = [72, 75, 78, 80, 82, 85]
result = stats.ttest_1samp(
    marks, popmean=75, alternative="two-sided"
)
print("t-statistic:", result.statistic)
print("p-value:", result.pvalue)
```

SciPy prints the t-statistic and p-value. Compare the p-value with a preselected alpha, such as 0.05. Avoid hard-coding a conclusion before the test runs.

Solution 3: Welch's independent t-test

```
from scipy import stats
a = [70, 72, 74, 76]
b = [80, 82, 84, 86]
result = stats.ttest_ind(a, b, equal_var=False)
print(result)
```

The sample means are 73 and 83. Welch's t-test evaluates evidence of a population mean difference, assuming the groups are independent and test conditions are reasonable.

Solution 4: p-value interpretation

Since 0.02 is below 0.05, reject H_0 at the 5% significance level. This is evidence against H_0 under the chosen test. It does not mean H_0 has a 2% probability of being true.

Solution 5: Paired t-test

```
from scipy import stats
before = [60, 65, 70, 72]
after = [66, 69, 75, 77]
result = stats.ttest_rel(before, after)
print(result)
```

The same students are measured twice, so the observations are matched.

Solution 6: Right-tailed alternative

```
H0:  $\mu = 75$ 
H1:  $\mu > 75$ 
```

The alternative is directional because the question asks whether the population mean is greater than 75.

Solution 7: Why data are paired

Each student's second score is related to that student's first score. Treating these observations as unrelated would ignore the matching and can estimate uncertainty incorrectly.

Solution 8: Type I and Type II errors

- Type I error: conclude that the method changes the population mean when the null hypothesis is true.
- Type II error: fail to detect a change when a real change exists.

Solution 9: Assumptions and checks

Examples include independence, influential outliers, strong non-normality in a small sample, and whether groups are independent or paired. Checks depend on the chosen test and study design.

Solution 10: Statistical vs practical significance

A large sample can detect a tiny difference with a small p-value. Practical importance depends on effect size, uncertainty, cost, educational relevance, and context. Report an effect size and confidence interval where possible.

12. Full project: Student Group Comparison Analyzer

Goal: compare two independent groups, calculate descriptive summaries, run Welch's t-test, and print a cautious conclusion. This is an educational example and does not establish causation.

Complete runnable project

```
import numpy as np
import pandas as pd
from scipy import stats

# Marks from two independent student groups
group_a = np.array([68, 72, 75, 77, 80, 82, 74, 76])
group_b = np.array([78, 81, 84, 86, 88, 79, 83, 85])

# Create a descriptive summary
summary = pd.DataFrame({
    "Group": ["Group A", "Group B"],
    "Sample Size": [len(group_a), len(group_b)],
    "Mean Marks": [np.mean(group_a), np.mean(group_b)],
    "Sample Std Dev": [
        np.std(group_a, ddof=1),
        np.std(group_b, ddof=1)
    ]
})

print("STUDENT GROUP SUMMARY")
print(summary.to_string(index=False, float_format=lambda x: f"{x:.2f}"))

# Welch's two-sided independent-samples t-test
result = stats.ttest_ind(
    group_a, group_b,
    equal_var=False,
    alternative="two-sided"
)

alpha = 0.05
mean_difference = np.mean(group_b) - np.mean(group_a)

print("\nHYPOTHESIS TEST")
print(f"Mean difference (B - A): {mean_difference:.2f}")
print(f"T-statistic: {result.statistic:.4f}")
print(f"P-value: {result.pvalue:.6f}")
print(f"Significance level: {alpha:.2f}")

if result.pvalue < alpha:
    print("Conclusion: Evidence of a difference between population group means at this significance level.")
else:
    print("Conclusion: Insufficient evidence of a difference between population group means at this significance level.")

print("\nINTERPRETATION REMINDER")
print("Statistical significance does not automatically imply a large")
print("or practically important difference, and does not prove causation.")
```

Line-by-line project walkthrough

- NumPy handles arrays, means, and standard deviations.
- Pandas builds a readable table of group summaries.
- SciPy supplies the inferential test.
- group_a and group_b contain marks from two separate groups.
- len(...) counts the observations in each group.
- np.mean(...) calculates each sample mean.
- np.std(..., ddof=1) calculates sample standard deviation using $n - 1$ in the denominator.
- pd.DataFrame({...}) organizes group labels and statistics into columns.
- summary.to_string(...) prints the table, formatting numbers to two decimal places.

- `stats.ttest_ind(...)` compares the independent groups' means.
- `equal_var=False` requests Welch's test, which allows unequal variances.
- `alternative='two-sided'` tests for a difference in either direction.
- `alpha = 0.05` defines the threshold before the decision.
- `mean_difference` calculates Group B's mean minus Group A's mean. A positive value means Group B's observed average is higher.
- `result.statistic` and `result.pvalue` provide the t-statistic and p-value.
- The `if/else` statement prints the decision based on the selected threshold.
- The final reminder warns that significance does not measure effect size or establish causation.

Expected output guide

There are 8 observations in each group. Group A's mean is 75.50, Group B's mean is 83.00, and the observed mean difference (B – A) is 7.50 marks. SciPy calculates the exact sample standard deviations, t-statistic, and p-value.

```
STUDENT GROUP SUMMARY
  Group Sample Size Mean Marks Sample Std Dev
Group A           8    75.50 [calculated]
Group B           8    83.00 [calculated]
```

```
HYPOTHESIS TEST
Mean difference (B - A): 7.50
T-statistic: [calculated by SciPy]
P-value: [calculated by SciPy]
Significance level: 0.05
Conclusion: [selected by p-value rule]
```

Bracketed values are placeholders rather than invented results. Run the code to see the exact output for your installed SciPy version.

Project extensions

- Accept marks entered by a user and validate numeric input.
- Plot each group's distribution and inspect potential outliers.
- Report the mean difference with a confidence interval.
- Calculate a standardized effect size and explain it in context.
- Save the summary using `summary.to_csv('group_summary.csv', index=False)`.
- Document how students were selected and whether the groups were independent.

13. Quick revision sheet

Question	Remember
What is H_0 ?	The null hypothesis, the baseline claim evaluated by a test.
What is H_1 ?	The alternative hypothesis describing the difference or direction.
What is alpha?	A preselected threshold, commonly 0.05.
What does $p < \alpha$ mean?	Reject H_0 under the chosen decision rule.
What does $p \geq \alpha$ mean?	Fail to reject H_0 ; not proof H_0 is true.
One sample vs reference?	<code>stats.ttest_1samp()</code> .
Two independent groups?	<code>stats.ttest_ind(..., equal_var=False)</code> for Welch's test.
Same subjects before and after?	<code>stats.ttest_rel()</code> .
Type I error?	Reject a true null hypothesis.
Type II error?	Fail to reject a false null hypothesis.

14. Completion checklist

- I can write H_0 and H_1 from a research question.
- I can explain alpha and a p-value accurately.
- I can distinguish two-tailed, right-tailed, and left-tailed tests.
- I can run one-sample, Welch's independent, and paired t-tests in SciPy.
- I can identify assumptions and limitations.
- I completed the Student Group Comparison Analyzer.

15. Next step

Day 22 can build on this foundation with correlation and covariance, which describe how two numerical variables vary together. Correlation measures association and does not, by itself, establish causation.

End of Day 21 • Hypothesis Testing in Data Science • Muhammad Ahmad